

## REICHENAU AN DER RAX, Austria

WMO: 113800

Lat: 47.6997N

Lon: 15.8369E

Elev: 1600

StdP: 13.87

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 10.5       | 15.0 | 5.5                           | 7.8 | 14.8 | 9.5 | 9.5 | 18.5 | 23.3                  | 33.3 | 21.4 | 34.9 | 2.4                   | 280  | 0.491 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.1                   | 85.9            | 67.9 | 82.4 | 66.6 | 79.3 | 65.0 | 69.7                | 81.9 | 68.1 | 79.4 | 66.5 | 76.8 | 4.5                  | 110  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 66.1                            | 102.3 | 73.2 | 64.2 | 95.5 | 71.8 | 62.7 | 90.3 | 70.9 | 34.7          | 81.5 | 33.3 | 79.7 | 32.0 | 76.9 | 77.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |
| 19.5              | 16.7 | 14.4 | 5.2                        | 91.7 | 5.4                | 3.2 | 1.3                                                | 94.0 | -1.9       | 95.9 | -4.9       | 97.7 | -8.8       | 100.1 |
|                   |      |      | 4.8                        | 73.1 | 5.4                | 2.0 | 0.9                                                | 74.6 | -2.2       | 75.8 | -5.2       | 76.9 | -9.1       | 78.4  |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |       |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |       |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |       |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |       |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |       |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |       |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |       |
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|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |       |
|                   |      |      |                            |      |                    |     |                                                    |      |            |      |            |      |            |       |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 49.3  | 31.8  | 34.9  | 40.7  | 48.9  | 56.3  | 62.8  | 66.0  | 65.4  | 57.4  | 50.0  | 41.9  | 34.0 | (6)  |
| (7)  |                                                                   | DBStd       | 13.86 | 8.52  | 8.64  | 7.66  | 7.08  | 6.10  | 6.04  | 5.45  | 5.48  | 5.96  | 6.71  | 7.54  | 8.21 | (7)  |
| (8)  |                                                                   | HDD50       | 2256  | 564   | 425   | 298   | 102   | 14    | 1     | 0     | 0     | 9     | 85    | 259   | 499  | (8)  |
| (9)  |                                                                   | HDD65       | 5964  | 1028  | 843   | 754   | 484   | 276   | 111   | 54    | 61    | 235   | 464   | 692   | 962  | (9)  |
| (10) |                                                                   | CDD50       | 1987  | 1     | 3     | 10    | 69    | 210   | 385   | 495   | 477   | 232   | 86    | 17    | 2    | (10) |
| (11) |                                                                   | CDD65       | 216   | 0     | 0     | 0     | 0     | 6     | 45    | 84    | 73    | 8     | 0     | 0     | 0    | (11) |
| (12) |                                                                   | CDH74       | 2245  | 0     | 0     | 0     | 14    | 117   | 478   | 804   | 728   | 98    | 6     | 0     | 0    | (12) |
| (13) |                                                                   | CDH80       | 586   | 0     | 0     | 0     | 0     | 12    | 123   | 219   | 220   | 12    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 4.8   | 5.7   | 5.8   | 5.6   | 5.3   | 4.8   | 4.5   | 4.7   | 3.9   | 3.9   | 3.9   | 4.7   | 5.2  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 42.2  | 2.0   | 2.0   | 3.0   | 2.6   | 4.6   | 5.6   | 5.4   | 4.8   | 4.0   | 3.0   | 2.8   | 2.4  | (15) |
| (16) |                                                                   | PrecMax     | 51.0  | 4.5   | 4.6   | 7.6   | 5.9   | 8.0   | 9.3   | 10.7  | 7.2   | 9.3   | 6.3   | 5.6   | 4.8  | (16) |
| (17) |                                                                   | PrecMin     | 30.4  | 0.3   | 0.3   | 1.1   | 0.2   | 1.7   | 2.4   | 1.1   | 1.4   | 1.7   | 0.5   | 0.0   | 0.4  | (17) |
| (18) |                                                                   | PrecStd     | 5.4   | 1.3   | 1.1   | 1.6   | 1.3   | 1.7   | 1.8   | 2.5   | 1.5   | 1.7   | 1.4   | 1.2   | 1.1  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 57.0  | 61.2  | 66.5  | 75.8  | 81.6  | 88.1  | 89.8  | 91.3  | 81.0  | 74.2  | 65.9  | 58.4 | (19) |
| (20) |                                                                   | MCWB        | 45.9  | 47.4  | 51.4  | 57.8  | 64.0  | 68.4  | 69.7  | 69.1  | 64.7  | 58.9  | 53.5  | 47.0  | (20) |      |
| (21) |                                                                   | 2%          | DB    | 52.4  | 55.9  | 61.7  | 71.0  | 77.3  | 83.5  | 85.5  | 85.7  | 76.7  | 69.3  | 61.9  | 53.8 | (21) |
| (22) |                                                                   | MCWB        | 44.0  | 45.3  | 49.2  | 55.3  | 61.7  | 67.4  | 68.1  | 67.9  | 63.2  | 57.4  | 52.1  | 45.5  | (22) |      |
| (23) |                                                                   | 5%          | DB    | 48.0  | 51.9  | 57.4  | 66.7  | 73.7  | 79.6  | 82.0  | 81.6  | 73.0  | 65.5  | 58.1  | 49.5 | (23) |
| (24) |                                                                   | MCWB        | 41.5  | 43.3  | 46.9  | 53.3  | 60.1  | 65.2  | 66.4  | 66.6  | 61.1  | 55.8  | 49.9  | 42.6  | (24) |      |
| (25) |                                                                   | 10%         | DB    | 44.2  | 47.9  | 53.5  | 62.8  | 69.4  | 75.6  | 78.5  | 77.6  | 69.5  | 61.9  | 54.3  | 45.6 | (25) |
| (26) |                                                                   | MCWB        | 38.8  | 41.0  | 45.4  | 51.3  | 57.7  | 63.4  | 65.3  | 64.6  | 59.5  | 54.0  | 47.6  | 40.1  | (26) |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 47.5  | 48.4  | 52.5  | 59.2  | 66.1  | 71.1  | 71.8  | 71.5  | 66.5  | 61.2  | 55.2  | 48.5 | (27) |
| (28) |                                                                   | MDCB        | 54.4  | 59.0  | 63.6  | 73.0  | 77.5  | 82.4  | 83.0  | 82.6  | 76.7  | 70.1  | 63.7  | 55.8  | (28) |      |
| (29) |                                                                   | 2%          | WB    | 44.6  | 45.8  | 49.9  | 56.4  | 63.2  | 68.5  | 69.8  | 69.4  | 64.4  | 58.7  | 52.8  | 46.0 | (29) |
| (30) |                                                                   | MDCB        | 50.9  | 54.4  | 60.0  | 68.7  | 73.9  | 80.4  | 82.0  | 81.8  | 73.9  | 66.5  | 60.3  | 52.1  | (30) |      |
| (31) |                                                                   | 5%          | WB    | 41.9  | 43.7  | 47.8  | 54.2  | 61.0  | 66.7  | 68.0  | 67.7  | 62.7  | 56.8  | 50.6  | 43.5 | (31) |
| (32) |                                                                   | MDCB        | 47.5  | 51.2  | 55.8  | 64.7  | 71.3  | 77.4  | 79.1  | 78.8  | 71.1  | 64.0  | 57.2  | 49.3  | (32) |      |
| (33) |                                                                   | 10%         | WB    | 39.4  | 41.3  | 45.9  | 52.3  | 58.9  | 64.5  | 66.3  | 65.9  | 60.5  | 54.9  | 48.2  | 40.4 | (33) |
| (34) |                                                                   | MDCB        | 43.7  | 47.7  | 52.6  | 61.2  | 68.0  | 73.3  | 76.2  | 75.9  | 67.7  | 61.0  | 53.3  | 45.1  | (34) |      |
| (35) | Mean Daily Temperature Range                                      |             | MDBR  | 12.8  | 14.3  | 15.8  | 18.7  | 18.7  | 18.7  | 19.1  | 20.0  | 17.3  | 17.6  | 13.4  | 12.6 | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 17.9  | 22.2  | 24.7  | 27.6  | 27.2  | 27.0  | 26.5  | 27.5  | 24.6  | 24.1  | 20.0  | 18.8 | (36) |
| (37) |                                                                   |             | MCWBR | 10.9  | 12.6  | 13.6  | 13.4  | 13.0  | 12.1  | 11.3  | 11.3  | 12.2  | 13.1  | 11.6  | 11.6 | (37) |
| (38) |                                                                   |             | MCDBR | 17.0  | 20.3  | 22.6  | 25.0  | 24.8  | 24.8  | 24.1  | 25.2  | 22.4  | 22.2  | 17.5  | 17.4 | (38) |
| (39) |                                                                   | 5% WB       | MCWBR | 10.9  | 11.9  | 12.9  | 12.5  | 12.6  | 11.9  | 11.0  | 11.2  | 11.9  | 12.7  | 10.8  | 11.5 | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.300 | 0.327 | 0.367 | 0.411 | 0.412 | 0.419 | 0.423 | 0.422 | 0.393 | 0.361 | 0.331 | 0.299 | (40) |      |
| (41) |                                                                   | taud        | 2.401 | 2.345 | 2.281 | 2.207 | 2.252 | 2.270 | 2.260 | 2.277 | 2.329 | 2.394 | 2.431 | 2.440 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 247   | 263   | 268   | 266   | 270   | 268   | 265   | 260   | 257   | 247   | 232   | 233   | (42) |      |
| (43) |                                                                   | Edh at Noon | 22    | 28    | 36    | 42    | 42    | 41    | 41    | 39    | 34    | 27    | 21    | 19    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 355   | 626   | 972   | 1358  | 1585  | 1729  | 1703  | 1520  | 1094  | 719   | 390   | 287   | (44) |      |
| (45) |                                                                   | RadStd      | 39    | 80    | 114   | 141   | 174   | 132   | 152   | 163   | 135   | 71    | 48    | 37    | (45) |      |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |     |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|-----|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |     |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m) |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |     |
| (47) | Regional (96 neighbors) | +0.99   | N/A    | N/A     | +1.53 | +0.81 | +0.72       | -205  | -272  | +158  | +88 |

Nomenclature: See separate page